

Week 4 Report — Core Implementation

Programme: IIMSTC – IoT & Embedded Systems (Cohort 4)

Team Size: 15 Members

Hardware Team	App Development Team	Software Team
Harsha Vardhan S	Krithi	Vijetha
Ananya Chandrashekar	Nagaraju M	S Vidyashree
Palaparthi Krishna Vamsi	Charan B S	Sachidananda M M
Durga Prasad S	Sachin Murchavar	—
Ramitha R	—	—
C Pavitra	—	—
Hadiya Khan	—	—
Abhinav P	—	—

Day 1 — Project Review & Implementation Preparation

The team conducted a detailed review of the progress achieved during the past week, focusing on system design, architecture, and the finalized list of components. Each module of the system was revisited to ensure clarity in functionality and integration. Documentation was refined and organized to create a clear reference for all team members. This step ensured alignment across hardware, software, and application teams, establishing a strong foundation before moving into the core implementation phase.

Day 2 — Planning & Component Finalization

A structured workflow was established by dividing responsibilities into specialized groups: hardware, software, app development, and documentation. Each group defined its scope, deliverables, and dependencies on other teams. Through collaborative discussions and evaluation of requirements, the components needed for the Smart Adaptive Cold Chain System were finalized, ensuring compatibility and efficiency. Additionally, a suitable location for offline implementation and testing was identified, considering factors such as accessibility, power availability, and safety.

Day 3 — Component Procurement & Initial Development

The procurement process for required hardware components was initiated, ensuring that all essential parts would be available for implementation without delay. Meanwhile, development activities progressed in parallel. The control logic for system operations began taking shape, focusing on core functionalities such as monitoring and response mechanisms. On the application side, the login interface was designed with attention to usability and security, while the initial framework of the dashboard was developed to display system data effectively.

Day 4 — Core Implementation & Debugging

The team moved into active system implementation, beginning with testing the basic design and verifying initial connections. The control logic was further refined to ensure accurate system responses and reliable performance under different conditions. Debugging played a key role during this phase, helping identify and resolve early-stage issues. Simultaneously, the application team completed the development of sign-up and login features, ensuring proper user authentication and smooth access to the system.

Day 5 — Circuit Configuration & Feature Development

The focus shifted towards stabilizing the system through continuous debugging and refinement. Hardware components were connected, and circuit configurations were finalized, including accurate pin mapping and secure connections. On the software and application side, new features were introduced to enhance usability and functionality. Data visualization through graphs was implemented, enabling users to interpret system data more effectively and gain meaningful insights from real-time monitoring.

Day 6 — Progress Review & Flow Analysis

A comprehensive review of the entire system was conducted to evaluate the progress achieved during the week. The team analyzed the overall implementation flow, identifying how different components interacted and where improvements could be made. Dependencies between hardware, software, and application layers were clearly understood, helping streamline future development. This session also helped define a structured roadmap for upcoming tasks, ensuring a more efficient and coordinated approach in the next phase.

Summary

Week 4 marked a critical transition from planning to execution. The team successfully initiated core implementation, aligned multiple system components, and addressed early-stage challenges through debugging and iterative development. By integrating hardware, software, and application modules, a strong and functional foundation was established, setting the stage for advanced features, optimization, and system scaling in the following weeks.